



SUSTAINED ATTENTION

Sustained attention is the ability to concentrate on a specific task, such as reading a book, for a long period of time. Brain imaging studies have shown that the frontal and parietal cortical areas, particularly in the right hemisphere of the brain, are associated with sustained attention.



SELECTIVE ATTENTION

Selective attention is the process of focusing intently on something specific, such as an object or sound, while tuning out our environment. Ignoring the sound of a car while paying attention to a phone is an example of selective attention.



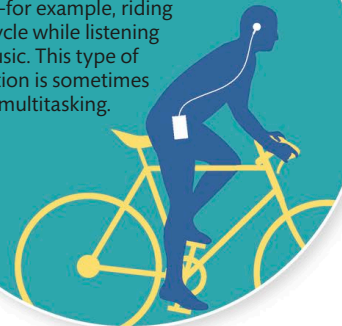
ALTERNATING ATTENTION

Alternating attention is the ability to switch attention quickly between tasks that require a very different cognitive response. Cooking dinner while periodically checking a recipe in a book is an example of alternating attention between different tasks.



DIVIDED ATTENTION

Divided attention is used so that we are able to perform two or more activities at the same time—for example, riding a bicycle while listening to music. This type of attention is sometimes called multitasking.



Types of attention

There are various types of attention, and the sort of attention that is required depends on the circumstances that we are in. Both sustained and selective attention are used when we need to focus fully on one stimulus. Alternating and divided attention are used when there are multiple inputs that we need to focus on at the same time. Attention is not an unlimited resource and the process of focusing our attention on something can be tiring, as it needs a significant amount of energy.

Distractions

The brain is not able to focus our attention constantly. Instead, it cycles rapidly between two different states: attention and distraction. During periods of distraction, the brain scans the environment to check that there is nothing more important to which it should be paying attention. This cycle is thought to give an evolutionary advantage to humans, allowing us to respond quickly to either new opportunities or threats.



During periods of distraction, brain scans environment

Looking for trouble

Even when we think we are focused on a task, our brain is checking the environment so that attention can be diverted if necessary.